

1 Methodology

This section details the complete processing pipeline used to extract, model and classify resting-state EEG signals from healthy controls (Typically developed) and first-episode psychosis patients.

1.1 Data Preprocessing

EEG recordings were acquired at a sampling rate of 256 Hz across 31 channels. The raw multi-channel signals were first smoothed using a Savitzky–Golay filter of order 3 and window length 13 to reduce high-frequency noise. Temporal derivatives were estimated from the smoothed signals to capture dynamic behaviour. Each channel was then standardised to zero mean and unit variance, and derivatives were scaled by the same factor to preserve dynamical consistency. Channels exhibiting negligible variance were discarded.

The continuous signals were segmented into overlapping windows of 1024 samples (≈ 4 s) with a stride of 512 samples. Within each window we computed a robust local centre as the per-channel median and obtained robust z-scores using the median absolute deviation (MAD). Time points whose dispersion exceeded a pre-defined threshold were treated as outliers and excluded. To ensure statistical stability, each retained window contained between 600 and 1000 valid samples.

1.2 From Dynamics to Simplicial Complexes

For every window we inferred pairwise and higher-order interactions among channels using sparse identification of nonlinear dynamics (SINDy). Given a state vector $X(t) \in \mathbb{R}^n$ and its time derivative $\dot{X}(t)$, we solved a sparse regression problem of the form

$$\dot{X}(t) \approx \Theta(X(t)) \Xi, \quad (1)$$

where $\Theta(\cdot)$ is a polynomial library of degree 2 and Ξ is a sparse coefficient matrix estimated via sequentially thresholded least squares with threshold parameter $\lambda = 0.2$. Linear monomials correspond to pairwise interactions (edges), while cross-terms involving three distinct channels correspond to higher-order interactions (triangles). Degenerate monomials such as x_i^2 were excluded from scoring.

We derived window-specific edge (1-simplex) and triangle (2-simplex) scores from the magnitude of Ξ . To obtain time-consistent topologies, we applied fixed global thresholds: the edge threshold τ_2 was chosen as the minimal bottleneck value of maximum spanning trees across all windows, ensuring connectivity, and the triangle threshold τ_3 was set to the 97.5th percentile of triangle scores. Each window thus yielded a closed two-dimensional simplicial complex comprising nodes (channels), edges and triangles. The corresponding cochain features were defined as the EEG sample at the window midpoint for 0-cochains, edge scores for 1-cochains and triangle scores for 2-cochains. These structures were exported in a format compatible with **TopoNetX**.

1.3 Temporal SCCNN–GRU Model

To integrate spatial topology with temporal dynamics, we designed a classifier combining a Simplicial Complex Convolutional Neural Network (SCCNN) with a Gated Recurrent Unit (GRU). For each window, the SCCNN took as input the cochains (H_0, H_1, H_2) and incidence matrices (B_1, B_2) and computed simplicial convolutions using the corresponding Laplacians $L_0, L_{1,d}, L_{1,u}, L_2$. This produced window-level embeddings capturing higher-order interactions. The sequence of embeddings was then fed into a GRU, which modelled temporal dependencies across windows. The final hidden state of the GRU was passed through a linear layer with softmax activation to produce a binary classification (healthy control vs. first-episode psychosis).

1.4 Training and Evaluation

The data were split on a per-subject basis into training, validation and test sets to avoid leakage across windows. Models were trained with the Adam optimiser (learning rate 10^{-3}) for 50 epochs using a cross-entropy loss. Early stopping based on validation accuracy was employed to prevent overfitting. Performance was assessed via accuracy and confusion matrices. All stages of the pipeline were orchestrated through a bash script to ensure reproducibility.